

Code-Layout, Readability and Reusability

Date	30 june 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in all modules
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders used for templates, static files and models
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve readability
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly represent their operations
5	Code Comments	Inline and block comments explain logic	Partial	Comments added for important sections only
6	Modular Design	Code is split into reusable functions/modules	Yes	Application divided into reusable modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused code snippets removed during testing
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Login Module	Python, Flask	User Authentication	High
2	Crop Prediction Model	Machine Learning, Python	Crop Recommendation System	High
3	Database Connectivity	SQLite / MySQL	User Data Management	Medium
4	Frontend Templates	HTML, CSS, Bootstrap	Multiple Web Pages	High
5	Input Validation Module	JavaScript, Python	Form Validation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder and file structure
Readability	4	Easy to understand with meaningful naming conventions
Reusability	5	Reusable modules and functions implemented effectively
Documentation / Comments	4	Basic comments provided for major functionalities
Overall Score	4.5	Good quality code with proper structure and maintainability